The Office of Dietary Supplements (ODS) of the National Institutes of Health (NIH)

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Probiotics

Fact Sheet for Consumers

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This is a general overview. For more in-depth information, see our health professional fact sheet.

What are probiotics and what do they do?

Probiotics are live microorganisms (such as bacteria and yeasts) that provide health benefits when they are taken in sufficient amounts. They are present in some fermented foods, added to other foods, and available as dietary supplements.

Probiotics act mainly in the digestive tract. Once in the digestive tract, probiotics may improve the function of your digestive system and other aspects of health.

Probiotic microorganisms are named by their genus, species, and strain. An example of a common probiotic microorganism is *Lacticaseibacillus rhamnosus* GG. In this case, *Lacticaseibacillus* is the genus, *rhamnosus* is the species, and GG is the strain. Other common probiotics include specific strains of *Limosilactobacillus reuteri* and *Saccharomyces boulardii*.

What foods provide probiotics?

Microorganisms are used to produce fermented foods, including yogurt, cheese, sourdough bread, pickles, apple cider vinegar, kombucha, miso, kimchi, and sauerkraut. Yogurt, for example, is made by adding live microorganisms (such as *Lactobacillus bulgaricus* and *Streptococcus thermophilus*) to milk. However, while microorganisms are used to make these foods, not all these foods necessarily contain probiotics with proven health benefits.

Manufacturers will sometimes add probiotics to unfermented foods, such as cereals, juices, milks, nutrition bars, smoothies, and infant and toddler formulas. If a food contains probiotics, the genus, species, and strain of the probiotic should be listed on the product label.

What kinds of probiotic dietary supplements are available?

Dietary supplements that are labeled as probiotics may contain different microorganisms, and the number of microorganisms they contain can vary widely between products. Because some of these supplements may not have been studied, any potential health effects they have may not be known.

The Supplement Facts label on a dietary supplement that contains probiotics lists the total weight of the microorganisms in the product. Many product labels also list the number of colony-forming units (CFUs) in a serving. The number of CFUs is a more accurate representation of how many active microorganisms are in a product than the total weight of the microorganisms. Examples of amounts of CFUs that you might see on a label are 1×10^9 (1 billion) CFUs and 1×10^{10} (10 billion) CFUs. Higher CFU counts alone do not necessarily mean that the product has greater health benefits. A product's health benefits, if any, are determined by both the specific microorganisms (or strains) it contains and the number of CFUs.

What are some possible effects of probiotics on health?

Scientists are studying probiotics to understand how they affect health. Here are some examples of what this research has shown.

Atopic dermatitis

Atopic dermatitis (eczema) is a skin condition that mostly affects children. People with atopic dermatitis have patches of dry, itchy skin. They also experience red rashes that can come and go, and these rashes may develop sores that ooze or bleed when scratched. Some studies have shown that when pregnant women take probiotics and their infants are given probiotics after birth, the infants might have a lower risk of developing atopic dermatitis and experience less severe symptoms of dermatitis. However, the effects vary depending on the probiotic strain used and whether the mother takes it during pregnancy, the infant is given it after birth, or both.

Pediatric acute infectious diarrhea

Acute infectious diarrhea in infants and children causes loose or liquid stools and three or more bowel movements within 24 hours. This condition is often caused by a viral infection and can last for up to a week. Some infants and children also develop fever and vomiting. Some studies have shown that certain probiotics shorten bouts of acute diarrhea by about 1 day. Other studies, however, have not shown that probiotics are effective.

Antibiotic-associated diarrhea

Antibiotics, such as erythromycin and penicillin, can kill beneficial microorganisms that live in the digestive tract, which sometimes results in diarrhea. Some probiotic strains might help reduce the risk of antibiotic-associated diarrhea in children and adults. This is especially true when people start taking these products within 2 days of the first antibiotic dose.

Necrotizing enterocolitis

Necrotizing enterocolitis (NEC) is a serious and sometimes deadly digestive tract illness that mostly affects preterm babies, especially those who are very small. Some probiotic strains might help reduce the risk of NEC in preterm infants. While these products are sometimes used in hospitals, additional

research on their safety and effectiveness is needed because the U.S. Food and Drug Administration (FDA) has raised some concerns about their use in preterm infants.

Inflammatory bowel disease

Inflammatory bowel disease (IBD) is a chronic disease that can affect children and adults. Types of IBD include ulcerative colitis and Crohn's disease. People with IBD commonly have diarrhea, stomach pain, or bloody stools that are caused by chronic inflammation in the digestive tract. Taking probiotics with medications might slightly reduce the severity of the symptoms of ulcerative colitis, but it does not seem to help people with Crohn's disease.

Irritable bowel syndrome

Irritable bowel syndrome (IBS) is a common disorder that can affect children and adults, especially women. It causes frequent stomach pain and discomfort, bloating, changes in bowel movement frequency, and diarrhea or constipation. The causes of IBS are unclear, but this condition may be a result of changes in the types of microorganisms in the digestive tract. Taking probiotics might help manage IBS symptoms. However, the effects vary depending on the probiotic strain used, how long it is used, and the symptom being treated.

Hypercholesterolemia

Very high levels of cholesterol in the blood (a condition known as hypercholesterolemia) and a buildup of cholesterol in blood vessel walls can block the flow of blood to the heart and increase the risk of cardiovascular disease. This condition can affect people of all ages, but it is most common in people over 40 years of age. Some studies have shown that probiotics can slightly lower the levels of total cholesterol and low-density lipoprotein (LDL) cholesterol, but other studies have found no benefits. More research is needed to understand the effects of probiotics on blood cholesterol.

Obesity

Some studies have shown that probiotics might slightly reduce body weight or body fat. Other studies have shown that probiotics have no effect or might even increase body weight. More research is needed to understand the effects of probiotics on body weight and body fat.

Can probiotics be harmful?

Most probiotics have been safely used in foods or dietary supplements for decades. Some of the same types of probiotics have been present in fermented food for thousands of years. In healthy people, probiotics may cause gas, but they rarely cause infections or other health problems.

However, probiotics might not be safe for everyone. The FDA has reported that probiotics might cause infections or even life-threatening illness in preterm infants. Probiotics might also cause problems such as bacterial infections in people who are already seriously ill or who have weak immune systems.

What should I know about choosing a probiotic?

There are no official recommendations that cover the use of probiotics by healthy people. If you want to try probiotics, ask your health care provider for advice about which probiotic to choose, what dose to take, and how long to use the product. On the product label, you can find information about the genus, species, and strain of the microorganisms; the expiration or use-by date; and storage instructions. Some probiotics need to be kept in the refrigerator, but others can be stored at room temperature.

The product label will also indicate the number of CFUs in the product. Some products list the number of CFUs at the time of manufacture whereas others list the number through the expiration date or use-by date. It is worth noting that the number of CFUs in a probiotic can decline over time. Therefore, the number of CFUs that a product contained when it was manufactured does not necessarily reflect the number of CFUs that the product contains when it is purchased or used. For this reason, some experts recommend choosing products that list the number of CFUs that the product will contain by the expiration date or use-by date instead of the number it contained at the time of manufacture.

Where can I find out more about probiotics?

- For more information on probiotics
  - Office of Dietary Supplements (ODS) Health Professional Fact Sheet on Probiotics (https://ods.od.nih.gov/factsheets/Probiotics-HealthProfessional/)
  - National Center for Complementary and Integrative Health, Probiotics (https://nccih.nih.gov/health/probiotics/introduction.htm)

    - United States Department of Agriculture (USDA) Agricultural Research Service, Keeping a Healthy Gut (https://www.ars.usda.gov/oc/utm/keeping-a-healthy-gut/)
- For more advice on buying dietary supplements
    - ODS Frequently Asked Questions: Which brand(s) of dietary supplements should I purchase? (https://ods.od.nih.gov/HealthInformation/ODS_Frequently_Asked_Questions.aspx#h9)
- For information on the government's food guidance system
    - Dietary Guidelines for Americans (https://www.dietaryguidelines.gov)
    - MyPlate (https://www.choosemyplate.gov/)

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